

Additional Technical Architecture Specification

Traffic Exchange Platform — Production Grade Expansion

1. API Contract Design

API Architecture

Type	Usage
REST API	CRUD / Configuration
WebSocket	Realtime analytics
Internal Queue	Worker communication

Base API Structure

```
/api/v1
├── auth
├── users
├── accounts
├── campaigns
├── analytics
├── workers
├── proxies
├── integrations
├── sessions
├── billing
├── admin
└── websocket
```

Authentication Endpoints

Method	Endpoint	Description
POST	/api/v1/auth/register	Register account
POST	/api/v1/auth/login	Login
POST	/api/v1/auth/logout	Logout
POST	/api/v1/auth/refresh	Refresh token
GET	/api/v1/auth/me	Current user

Campaign Endpoints

Method	Endpoint	Description
GET	/api/v1/campaigns	List campaigns
POST	/api/v1/campaigns	Create campaign
GET	/api/v1/campaigns/:id	Campaign detail
PUT	/api/v1/campaigns/:id	Update campaign
DELETE	/api/v1/campaigns/:id	Delete campaign
POST	/api/v1/campaigns/:id/start	Start campaign
POST	/api/v1/campaigns/:id/stop	Stop campaign
POST	/api/v1/campaigns/:id/pause	Pause campaign

Proxy Endpoints

Method	Endpoint	Description
GET	/api/v1/proxies	List proxy
POST	/api/v1/proxies	Add proxy

DELETE	/api/v1/proxies/:id	Remove proxy
POST	/api/v1/proxies/test	Test proxy
POST	/api/v1/proxies/rotate	Rotate proxy

Worker Endpoints

Method	Endpoint	Description
GET	/api/v1/workers	Worker list
GET	/api/v1/workers/status	Worker stats
POST	/api/v1/workers/restart	Restart worker
GET	/api/v1/workers/health	Health check

Analytics Endpoints

Method	Endpoint	Description
GET	/api/v1/analytics/overview	Global stats
GET	/api/v1/analytics/realtime	Live analytics
GET	/api/v1/analytics/campaign/:id	Campaign analytics
GET	/api/v1/analytics/geo	GEO analytics

WebSocket Channels

analytics:update
campaign:update
worker:update
proxy:update
queue:update
system:health

Standard API Response

Success Response

```
{  
  "success": true,  
  "message": "Campaign created",  
  "data": {}  
}
```

Error Response

```
{  
  "success": false,  
  "error": {  
    "code": "CAMPAIGN_LIMIT",  
    "message": "Campaign limit exceeded"  
  }  
}
```

2. Database Schema Detail

Database Architecture

Main Tables

users
roles
subscriptions
campaigns
campaign_targets
campaign_sessions
browser_sessions
proxy_pools
proxy_logs
fingerprints
worker_nodes
worker_logs
analytics_events

traffic_logs
integrations
billing_transactions
credit_logs
system_logs

users

Column	Type	Index
id	uuid	PK
email	varchar	UNIQUE
password_hash	text	
role_id	uuid	INDEX
is_active	boolean	
created_at	timestamp	INDEX

campaigns

Column	Type	Index
id	uuid	PK
user_id	uuid	INDEX
name	varchar	
target_url	text	
status	enum	INDEX
daily_limit	int	
geo_mode	enum	
created_at	timestamp	INDEX

proxy_pools

Column	Type	Index
id	uuid	PK
user_id	uuid	INDEX
type	enum	
host	varchar	
port	int	
username	varchar	
password	text	
country	varchar	INDEX
status	enum	

browser_sessions

Column	Type	Index
id	uuid	PK
campaign_id	uuid	INDEX
proxy_id	uuid	INDEX
fingerprint_id	uuid	INDEX
worker_id	uuid	INDEX
status	enum	
started_at	timestamp	INDEX

analytics_events

Column	Type	Index
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id	uuid	PK
campaign_id	uuid	INDEX
session_id	uuid	INDEX
event_type	varchar	
country	varchar	INDEX
duration	int	
created_at	timestamp	INDEX

Recommended Index Strategy

Composite Index

campaign_id + created_at
worker_id + status
proxy_id + status
user_id + created_at

Recommended Partitioning

Partition Tables

- analytics_events
- traffic_logs
- worker_logs

Partition:

- monthly
-

3. Worker Concurrency Policy

Browser Resource Estimation

Resource	Estimated Usage
Chromium Instance	300-500MB
Browser Context	50-100MB
Playwright Worker	1-2 CPU

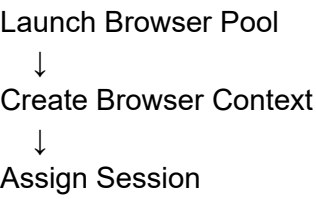
Recommended Limits

VPS Spec	Browser Limit
4 CPU / 8GB	10 browser
8 CPU / 16GB	20 browser
16 CPU / 32GB	40 browser

Worker Concurrency Rules

Rule	Value
Max browser per worker	5
Max concurrent session	20
Max RAM usage	80%
Restart threshold	85%
Browser timeout	10 minutes

Browser Lifecycle



↓
Track Usage
↓
Recycle Context

Session Persistence Policy

Session Type	Persistence
Sticky session	30 minutes
Persistent profile	7 days
Ephemeral session	one-time

4. Retry Strategy & Error Handling

Queue Retry Policy

Queue	Retry Count
campaign_queue	3
analytics_queue	5
proxy_queue	2
worker_queue	3

Retry Backoff Strategy

Exponential Backoff

1s
2s

5s
10s
30s

Failure Handling

Error Type	Action
Browser crash	restart browser
Proxy timeout	rotate proxy
CAPTCHA	mark failed
403 forbidden	cooldown
Worker dead	reassign job

Dead Letter Queue

Store:

- job payload
 - worker id
 - proxy info
 - stack trace
 - retry history
-

Worker Health Monitoring

Health Metrics

Metric	Threshold
RAM usage	80%
CPU usage	85%

Browser crash rate	10%
Queue lag	30s

5. RBAC (Role Based Access Control)

Role Structure

Role	Access
user	own campaigns
moderator	monitor users
admin	system management
superadmin	infrastructure

Permission Matrix

Permission	User	Mod	Admin
create_campaign	yes	yes	yes
delete_campaign	own	yes	yes
manage_users	no	limited	yes
restart_worker	no	yes	yes
system_settings	no	no	yes
billing_access	own	no	yes

RBAC Middleware Flow

Request
↓
JWT Validation
↓
Role Validation
↓
Permission Check
↓
API Access

Security Middleware

Middleware	Purpose
JWT	authentication
Rate Limit	anti abuse
Helmet	secure headers
Audit Log	activity tracking
CSRF	request validation

Audit Logging

Track:

- login activity
 - campaign changes
 - billing activity
 - worker actions
 - admin actions
-

6. Billing & Credit Logic

Billing Model

Credit-Based System

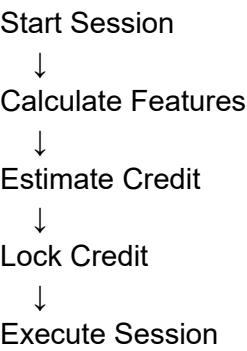
Credit Usage Rules

Action	Credit
1 standard session	1
Residential proxy	+2
Mobile proxy	+5
GEO targeting	+1
Advanced stealth	+1
Session persistence	+1

Subscription Plans

Plan	Credits
Free	100/day
Starter	10k/month
Pro	100k/month
Enterprise	custom

Credit Deduction Flow



↓
Finalize Usage

Billing Tables

subscriptions

Column	Type
id	uuid
user_id	uuid
plan	enum
credit_limit	bigint
expired_at	timestamp

credit_logs

Column	Type
id	uuid
user_id	uuid
amount	int
type	enum
source	varchar
created_at	timestamp

Recommended Payment Gateway

Indonesia

- Midtrans
- Xendit

Global

- Stripe
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Recommended Initial Infrastructure

MVP Deployment

Nuxt App
Redis
PostgreSQL
Worker Service
Cloudinary

Scaling Ready Architecture

Nuxt App
↓
Redis Queue
↓
Distributed Workers
↓
Analytics Pipeline

Final Engineering Recommendation

Phase 1 Goal

Focus:

- stable worker
- stable queue

- stealth quality
- proxy management
- realtime analytics

Avoid:

- premature microservices
 - Kubernetes complexity
 - Kafka complexity
 - Rust services too early
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